PAPER_200: Um Universal Magnetism Taxonomy — Complete Variant Catalogue

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi GMc}{\kappa_{\text{es}}}\left(1 - [SSq]\cdot e^{-\kappa\,\Delta t}\right), \quad [SSq] = 0.57$$

Abstract

Universal Magnetism (Um) in UQFF is a general magnetic energy density operator that scales with vacuum density ?_vac and an exponential damping factor (1-e^{-?t}). This paper catalogs all named Um sub-variants extracted from the BB_C_Equations_04Sept2025.pdf, covering 50+ unique astrophysical and cosmological regimes: cosmological magnetism, reheating, BBN, MHD dynamo, quasi-normal modes, Blandford-Znajek, photo-evaporation, planet migration, terminal velocity, Press-Schechter, star formation efficiency, BH entropy, evaporation, angular momentum, jet velocity, GW chirp mass, QNM, duty cycle, peculiar velocity, ionization, deuterium, Friedmann-1, QNM damping, Arnett, reheating, Kazantsev dynamo, Alfvén Mach, field reversal, and more.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Universal Magnetism General Form

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Um,X(arg) = [S_j µ_j(t, ?_vac,[SCm])/r_j] × (1 - e^{-?t} · cos(pt_n)) × F_X

or (single j variant):
Um,X(arg) = µ(?_vac) × (1 - e^{-?t}) × F_X

where:
  µ_j = magnetic moment density at scale j
  ?_vac,[SCm] = superconductive vacuum density
  ? = exponential decay rate
  t_n = normalized time: t_n = t/t_Hubble·(1+H(z)·t0)
  F_X = system-specific magnetic function
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2. Cosmological Magnetism Variants

Symbol	F_X Expression	Physical Context
Um,cosmo(t,a)	$(k_c^2/a^2)/H^2$	UQFF cosmological magnetic field scaling
Um,reh(N)	$(30V_{end}/(p^2 g_*))^{\{1/4\}\cdot e}\{-3N_{reh}/4\}$	Reheating post-inflation
Um,BBN(t)	v(3/(32pG?_rad)) ˜ 180 s (T~0.1 MeV)	Big Bang Nucleosynthesis
Um,fried1(a)	$O_m(1+z)^3 + O_k(1+z)^2 + O_?$	Friedmann-1 Hubble term
Um,wde(a)	? ? a^{-3(1+w)}	Dark energy equation of state
Um,deden(a)	w(a) = w0 + w_a(1-a) (CPL form)	Dark energy density evolution
Um,grow(a)	Growing mode D ? a (matter era), suppressed by DE	Structure growth factor
Um,curv(?)	?? = v(2e)·H·M_pl	Curvature perturbation from inflation
Um,ng(f)	From d? curvature on superhorizon scales	Non-Gaussianity

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3. Black Hole Thermodynamics Variants

Symbol	F_X Expression	Physical Context
Um,haw(M)	T = ?/(2p) ; ? = c4/(4GM)	Hawking temperature surface gravity
Um,bhent(M)	Holographic S ? A/l_{pl}^2	Bekenstein-Hawking entropy area law
Um,evapl(M)	$dM/dt = -P/c^2$	Black hole evaporation mass loss rate
Um,qnm(a)	$a\cdot c^3/(2GM) \times [l=2, m=2]$	QNM ringdown frequency (Berti fits)
Um,damp(a)	Q ˜ 10 for l=2, m=2	QNM quality factor (damping)
Um,bz(a)	Power ? $B^2 O^2_H \cdot R4_H$ (electromagnetic extraction)	Blandford-Znajek energy extraction
Um,bz2(a)	$?/(16p)F^2_{BH}\cdot O^2_{BH}/c$	BZ jet power (EHT-updated form)

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4. Compact Object and Stellar Variants

Symbol	F_X Expression	Physical Context
Um,MHD	$4p?\cdot v^2_{turb}\cdot M_A$	MHD turbulent Alfvén scaling
Um,termv(G)	L/L_Edd ˜ 1 ; v_8 = $v(2(L/?c)-v^2_{esc})$	Terminal velocity in radiation winds
Um,sfe(z)	?_* ? $M^{\{1.1\}\cdot(1+z)}\{2.25\}$	Star formation efficiency evolution
Um,duty(t)	exp(-t/t_cool)	AGN feedback duty cycle
Um,ang(t,r)	$v(GM/r^3)\cdot r_A$; accretion adds $L=?r^2O$	Angular momentum accretion

Symbol	F_X Expression	Physical Context
Um,jetvel(r)	O·r_A·(r_A/r0)^{1/2}	Jet velocity from Alfvén radius
Um,glitch(t)	Angular momentum transfer ?L = I_s·?O	Superfluid vortex glitch
Um,ms(M)	Eddington: L ˜ µ4M³ (opacity, composition µ)	Main-sequence L-M relation
Um,ml(M)	Calibrate from HR diagram	Mass-luminosity relation
Um,relax(?)	s_v³/(G²?m)	Two-body relaxation time
Um,evap(N)	Integrate exponential cluster dissolution	Star cluster evaporation
Um,star(N)	136·t_relax/ln(0.02N)	N-body cluster relaxation timescale

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5. Gravitational Wave Variants

Symbol	F_X Expression	Physical Context
Um,chirp(f)	(32/5)(G??$^{\{5/3\}/c5)(pf)}${10/3}	GW chirp mass inspiral energy loss
Um,orbdec(e)	dE/dt = -L, E = -GM·µ/(2a)	Orbital decay by GW quadrupole radiation
Um,peri(a)	Kepler: a³/P² = GM/(4p²)	Post-Keplerian periastron advance
Um,kilo(t)	Diffusion t_d² = 3?M/(4pcv²)	Kilonova light curve peak

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6. ISM and Plasma Variants

Symbol	F_X Expression	Physical Context
Um,malf(B)	M_A <1 sub-Alfvénic (kinetic ˜ B²/(8p))	Alfvén Mach number turbulence regime
Um,rev(?)	?B/?t = ?×(aB - ??×B) (growth vs diffusion)	Magnetic field reversal (parity)
Um,dyn(k)	?E dk ; E_M = ?E·dk (Kazantsev E spectrum)	Magnetic energy growth via dynamo
Um,alf(B)	Wave speed from linearization ?²b/?t² = v_A²·?²b/?z²	Alfvén wave propagation
Um,turb(l)	?v²/t_eddy ? l^{2/3} (Kolmogorov-like ISM)	Turbulent energy cascade rate
Um,cshock(z)	(?×B)×B/(4p) + ?_i·?_in·(v_n-v_i)	C-type shock (magnetized, continuous)
Um,jshock(M)	v1/c_s × T2 ? v_s²	J-type shock (discontinuous) Rankine-Hugoniot
Um,dsa(p)	B?¹·pc/e × ? ? B²	Diffusive shock acceleration spectrum

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7. Cosmological Structure Formation Variants

Symbol	F_X Expression	Physical Context
Um,ps(M)	s(M) rms fluctuation ? d ln s/d ln M	Press-Schechter halo mass function
Um,halo(z)	?P(k)W^2(kR)d^3k/$(2p)^3$ × (1+z)^{m/2}	Extended Press-Schechter halo mergers
Um,mig(a)	da/dt = 2G/(M_pv(GM_*·a))	Type-I planet migration
Um,migr(a)	S(H/r)$?^3$	Disk migration timescale
Um,acc(t)	Accretion ? = L/(ec^2)	BH accretion (Eddington-limited)
Um,disk(?)	?_* ? M$^{\{3/2\}_gas/R}${3/2}	Star formation rate in disk
Um,burst(t)	Enhancement 10–100× at low z	Merger-induced starburst enhancement
Um,metal(Z)	Z = Y·SFR/?_gas - Z·?_out/M_gas	Metal enrichment mass balance
Um,sfe(z)	Efficiency limited by cooling/feedback	Star formation efficiency
Um,fb(?)	?_*·SFR (fraction ?)	Feedback energy injection (SN/AGN)
Um,bhmf(z)	erfc(d_c(z)/v(2)s(M′,z)) × extend to M_f	BH mass function (EPS)

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8. Reionization and Early Universe Variants

Symbol	F_X Expression	Physical Context
Um,ion(t)	?a_B·n^2_e·C·dt	Ionization fraction integrated
Um,reion(t)	?a_B·n^2_e·C·dt (full Madau model)	Cosmic reionization
Um,eta(t)	Fit networks to D, He abundances	Baryon-to-photon ratio ?
Um,deb(T)	Weak freeze at T~1 MeV; D photodissociation below	Deuterium bottleneck onset
Um,recomb(z)	Optical depth t = ?s_T·n_e·dl	Recombination epoch
Um,cmb(k)	Transfer ?_l^T: Sachs-Wolfe + acoustic oscillations	CMB power spectrum
Um,bub(t)	Expand for moving ionization front	Bubble growth during reionization

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9. Dark Matter and Cluster Variants

Symbol	F_X Expression	Physical Context
Um,nfw(r)	Fit to universal NFW form ?_s/(x·$(1+x)^2$)	NFW dark matter halo profile
Um,nfwrot(x)	Flat rotation curve for r » r_s	NFW rotation curve
Um,sidm(t)	Exponential density flattening (G·t ˜ 1)	SIDM core formation
Um,vir(r)	s^2_v = GM/(3r) (spherical approximation)	Virial theorem mass estimation

Symbol	F_X Expression	Physical Context
Um,virx(r)	Matches Chandra cluster observations (e.g., M4)	X-ray binary virial trace
Um,mach(?)	Matches Coma radio relic shocks (M~2–3)	Merger shock Mach number
Um,merg(t)	$3r_{vir}/(5GM)$	Virial merger crossing time
Um,cross(M)	Matches ~2 Gyr for relaxed clusters	Merger timescale (dynamical age)
Um,heat(T)	Balance with duty-cycled AGN	Cool core AGN heating rate
Um,whim(M)	µ ˜ 0.6 (mean molecular weight)	WHIM temperature from virialization
Um,cool(T)	t = E/E	Cooling time (bremsstrahlung T > 107 K)
Um,lens(?)	$?_E^2$ = a·? ? Einstein radius	Strong gravitational lensing

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10. Additional Specialized Variants

Symbol	F_X Expression	Physical Context
Um,conv(T)	Velocity from ?g dz ˜ v^2	Convective turnover (mixing length)
Um,lqcf(?)	Friedmann from bounce dynamics	LQC effective Friedmann
Um,lqcp(k)	Power tilt + UV suppression	LQC perturbation pre-bounce
Um,bounc(?)	? ? $1/(G \cdot t^2)$ singularity capped	LQC bounce density cap
Um,holoent(?)	Area(?_A)/(4Gh/c^3)	Holographic entanglement entropy (Ryu-Takayanagi)
Um,pec(r)	Spherical void: integrate Poisson	Peculiar velocity in voids
Um,voidden(a)	d ? a?^{1} in matter domination	Void underdensity evolution
Um,jeans(T)	Perturb: $?^2 = c^2\_s \cdot k^2$ - 4pG?	Jeans instability dispersion
Um,fermi(E)	Average $(?E/E)^2/2$ per scatter	Fermi-II acceleration (2nd order)
Um,knee(B)	Balance t_acc = t_age ˜ R/u_s	Cosmic ray knee from DSA limit
Um,diff(E)	Fit to secondaries (B/C ratio)	Cosmic ray diffusion coefficient
Um,esc(F)	Adjust for penetration K(?)	Photoevaporative escape (exoplanet)
Um,rv(e)	Orbital v_p = 2pa/P·sin(i)	Radial velocity exoplanet detection
Um,upar(r)	U = G/(n_H·c) (ionization parameter)	AGN photoionization parameter
Um,coup(?)	Couple fraction to kinetic energy	AGN feedback thermal coupling
Um,photo(t)	? ? $F\_X \cdot R^3\_p/(GM\_p/R\_p)$	Photoevaporative mass loss rate
Um,puls(t)	Torque I·dO/dt = -L/O	Pulsar spin-down magnetic torque
Um,tov(r)	Add GR: P/c^2 energy density, $(1-2GM/rc^2)$	TOV stellar structure GR correction
Um,reljet(z)	Approximate sigmoid for gradual G rise	Relativistic jet velocity profile
Um,after(t)	N(E) ? E^{-p} × t^{-1.2}	GRB afterglow synchrotron spectrum
Um,fire(dt)	Internal shocks at dr ˜ $c \cdot dt/G^2$	GRB fireball prompt emission

Symbol	F_X Expression	Physical Context
Um,sedov(t)	Integrate momentum: d/dt(Mv)=0	Sedov-Taylor blast wave

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11. Summary Statistics

Metric	Value
Total named Um variants catalogued	55+
Document source	BB_C_Equations_04Sept2025.pdf (177 pages)
Total equation pairs (F_UBii + Um) in source	~1,350+ unique positions
Scale coverage	10?^{3}4 N·m (molecular rotors) ? 10^{2}7 m (D_universe)
Q_wave std	6.35×104 J/m^{3} (47-scale average)

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12. References

- `grok_share_7514fe.txt` lines 2676–2762 (first Um catalogue from BB_C_Equations_04Sept202
- `grok_share_7514fe.txt` lines 6001–6380 (second/final Um catalogue)
- PAPER_198: F_UBii Taxonomy Part 1 (Compact Objects)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological/Dark Sector)
- PAPER_196: Triadic Master Equation System

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13. Operational Implementation Status

The following table classifies each Um variant by its production status in the current codebase (`MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, `CondensedPhysics2.py`).

Status	Meaning
□ **Operational**	Fully implemented, returns numerical result for any input dataset
□ **Partial**	Core formula implemented; specialized sub-terms use placeholder values
□ **Reference**	Equation documented for reproducibility; not yet callable from pipeline

Core Um Variants

Symbol	Description	Status	Notes
Um (base)	`Σ_j[μ_j/r_j·(1−e^{−γt·cos(πt_n)})·φ̂_j]·P_SCm·E_react`	□ **Operational**	`compute_Um_SOURCE4`, `compute_Um()`
Um (Heaviside-amplified)	Um_base × (1+10^{13}·Θ(ρ_SCm−ρ_c)) × (1+A_q·cos(Δω·t))	□ **Operational**	PAPER_421 — integrated v4.75

Symbol	Description	Status	Notes
Um,BZ	Blandford-Znajek power extraction	□ **Operational**	`CondensedPhysics2.py` BZ class
Um,haw	Hawking temperature surface gravity	□ **Operational**	`bh_thermodynamics_modul`
Um,qnm	QNM ringdown frequency	□ **Operational**	`CondensedPhysics.py` QNM class

Reference-Only Variants (50+)

All remaining Um,X variants catalogued in §2–§10 of this paper are □ **Reference** status. They contain correct analytical expressions derived from BB_C_Equations_04Sept2025.pdf but are not yet wired into the main computation pipeline.

Operational upgrade path: To promote any variant from Reference → Operational, implement a calculator class in `CondensedPhysics.py` following the `compute_Um()` interface pattern, then register it in the `PhysicsTermRegistry`.

> *Operational status assessed: v4.75 (January 28, 2026 integration).*
> *Um Heaviside amplifier + quasi-periodic modifier now operational (PAPER_421).*